Session 3

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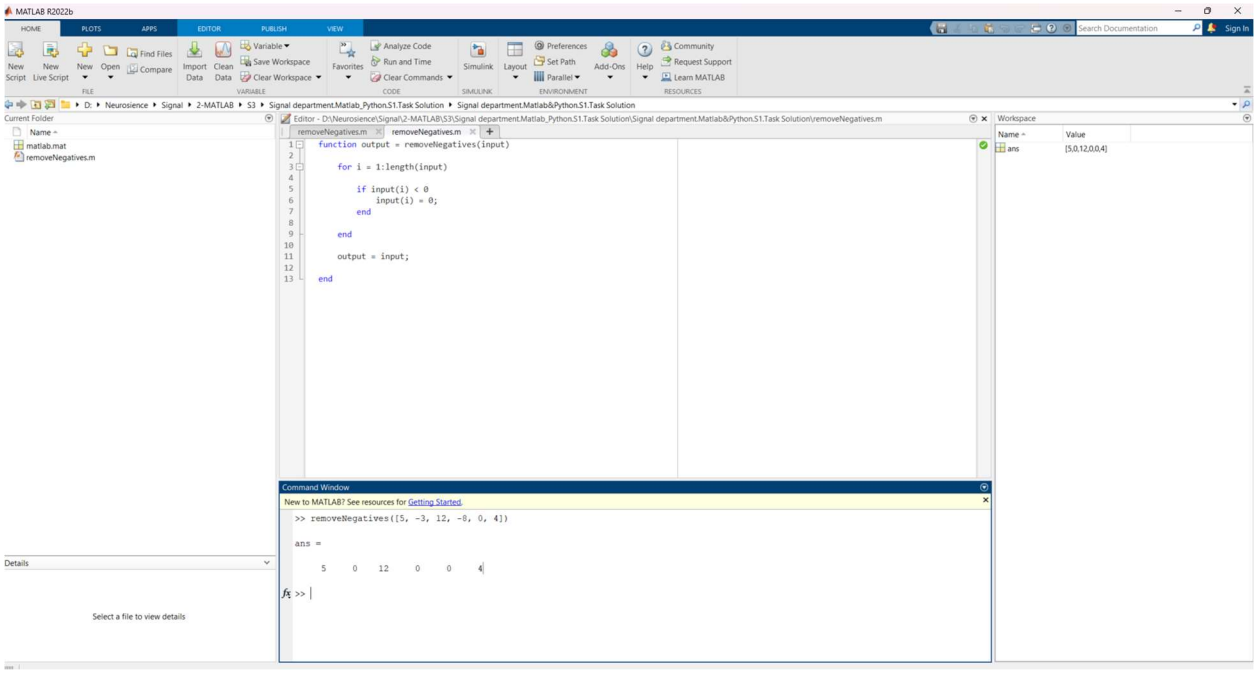
Please complete the following steps for each task:

1. Write the requested MATLAB function in the MATLAB Editor and take a clear screenshot of your code.
2. Run/test your function in the Command Window (or in a test script) using the provided example input, and take a screenshot of the function call and its resulting output.
3. Paste both screenshots below each task.

Task 1:

Write a function named removeNegatives that takes a vector as input. Using a for loop and an if condition, replace every negative number with zero and return the modified vector.

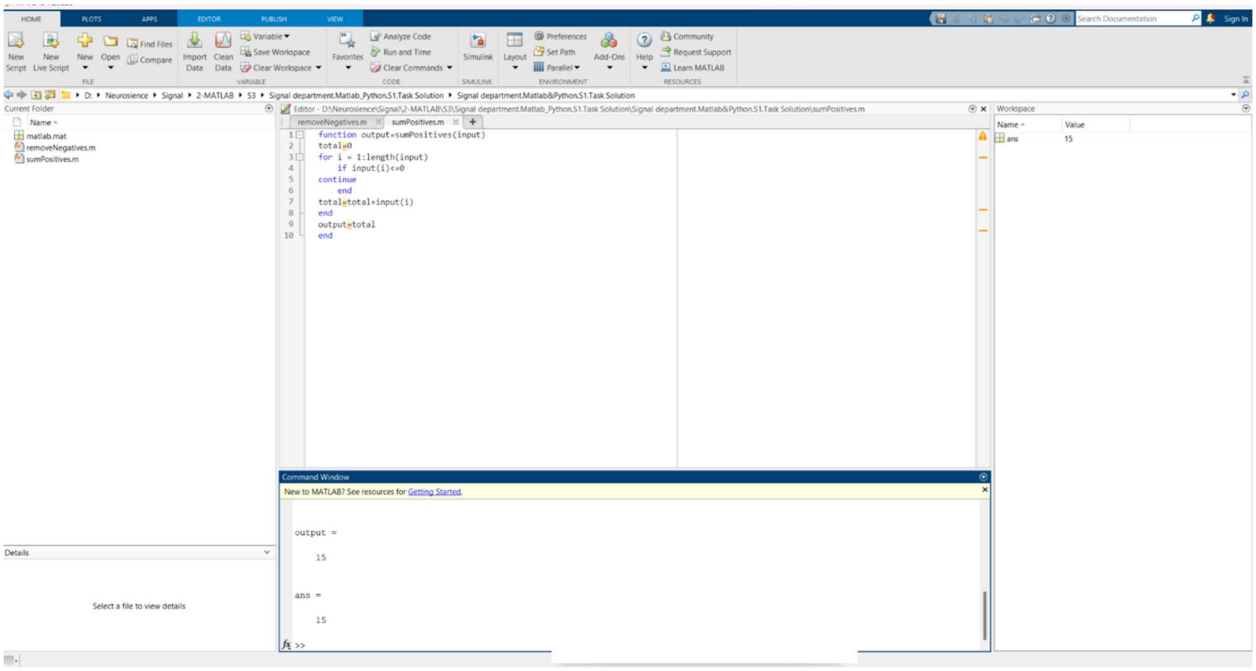
Example Input: [5, -3, 12, -8, 0, 4]



Task 2:

Write a function named sumPositives that calculates the sum of all strictly positive numbers in a vector. If a number is negative or zero, skip it using continue.

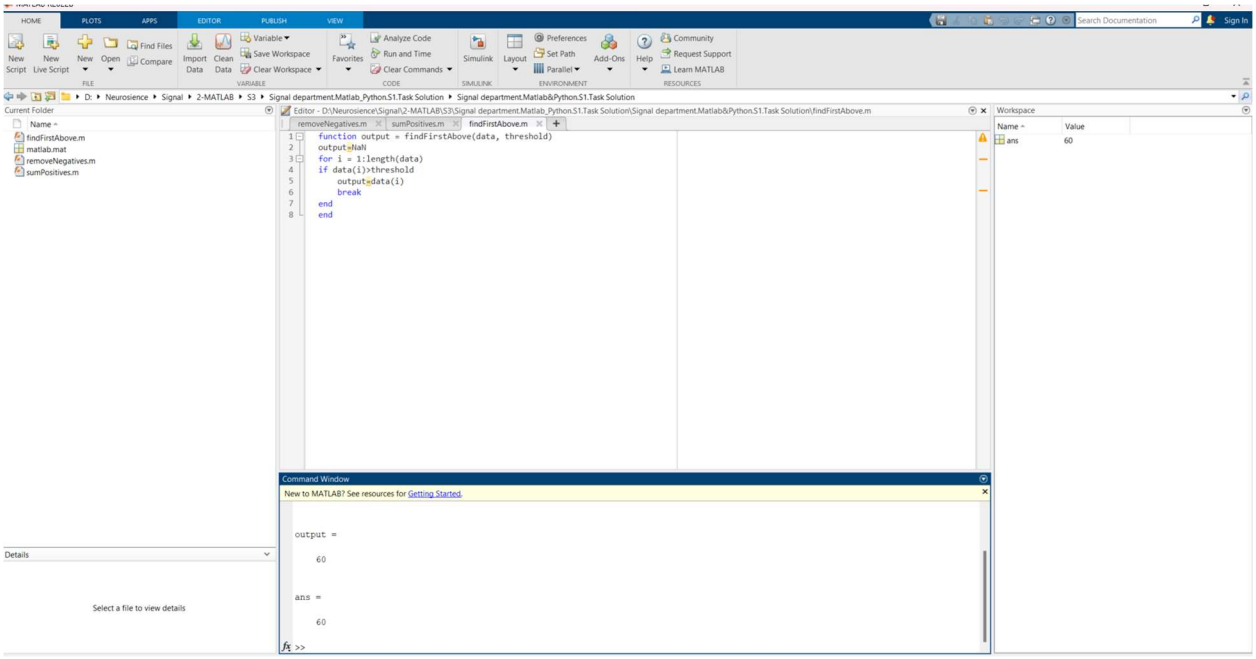
Example Input: [4, -10, 6, 0, 5, -2]



Task 3:

Write a function named findFirstAbove that takes a vector and a threshold value. Loop through the vector, find the first element strictly greater than the threshold, stop immediately using break, and return that value. If none exists, return NaN.

Example Input: data = [10, 25, 60, 80, 40]; threshold = 50;

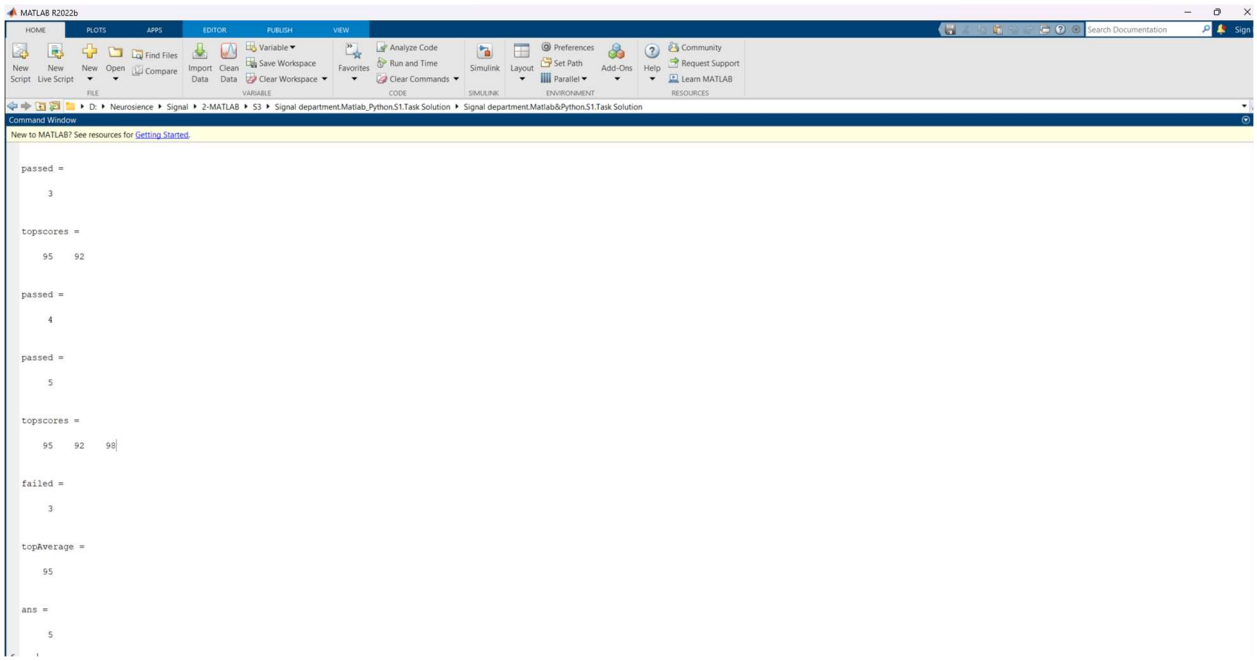


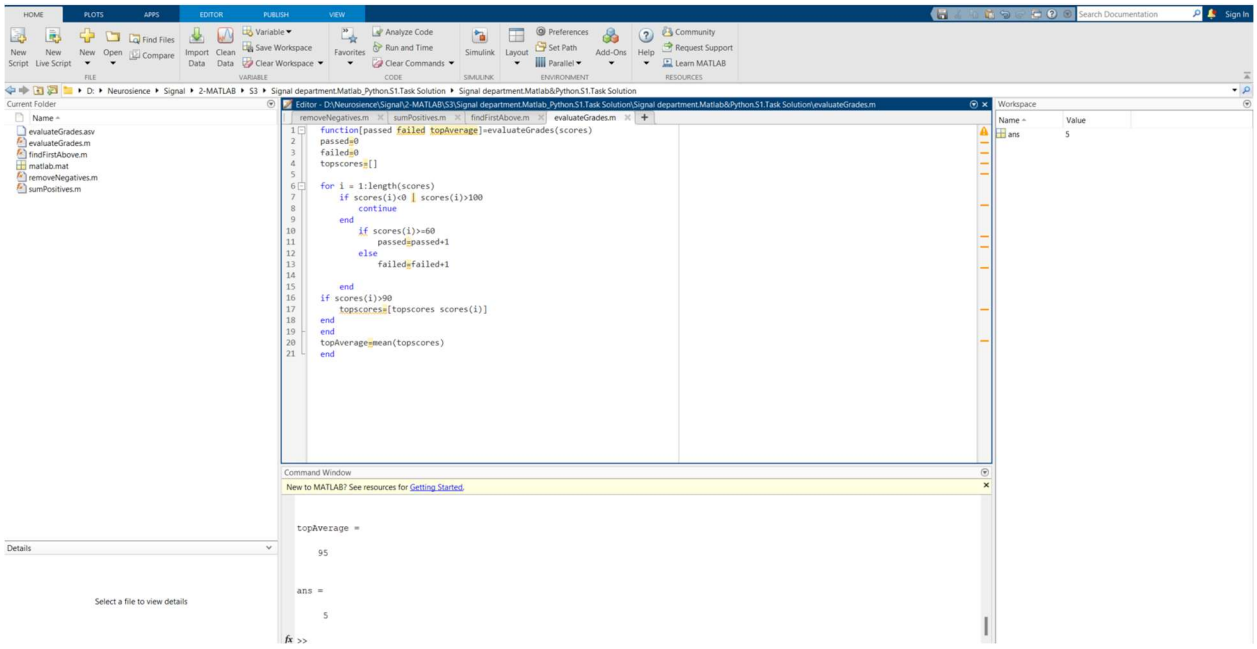
Task 4:

Write a MATLAB function named evaluateGrades that takes a vector of student grades as input and performs the following tasks using a for loop and conditional statements (if-else / continue):

1. Count the number of students who passed (grade ≥ 60) and failed (grade < 60).
2. Identify grades greater than 90and calculate the average (mean) score of these top-performing students.
3. If an invalid score is encountered (grade < 0 or grade > 100), skip it using the continue statement.

Example Input: scores = [45, 85, 95, 110, -5, 58, 92, 74, 98, 30])


MATLAB R2022b
Command Window
New to MATLAB? See resources for Getting Started.
passed =
3
topscores =
95 92
passed =
4
passed =
5
topscores =
95 92 98
failed =
3
topAverage =
95
ans =
5


Editor - D:\Neurosience\Signal\2-MATLAB\S3\Signal department.Matlab_Python.S1.Task Solution\Signal department.Matlab&Python.S1.Task Solution\evaluateGrades.m
removeNegatives.m
sumPositives.m
findFirstAbove.m
evaluateGrades.m
function[passed failed topAverage]=evaluateGrades(scores)
passed=0
failed=0
topscores=[]
for i = 1:length(scores)
if scores(i)<0 | scores(i)>100
continue
end
if scores(i)>=60
passed=passed+1
else
failed=failed+1
end
if scores(i)>90
topscores=[topscores scores(i)]
end
end
topAverage=mean(topscores)
end
Current Folder
evaluateGrades.asv
evaluateGrades.m
findFirstAbove.m
matlab.mat
removeNegatives.m
sumPositives.m
Workspace
ans 5
Command Window
New to MATLAB? See resources for Getting Started.
topAverage =
95
ans =
5
Details
Select a file to view details